

Industrial DIGITAL THERMOMETER

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Presented here is a 4-digit thermometer based on ATmega16 microcontroller (MCU), which can measure temperature up to 1024°C.

Normal thermometers have a limited range, and can easily get damaged at very high temperatures. So thermocouples are used in industries for measuring very high temperatures. This circuit uses a K-type thermocouple along with MAX6675 IC for temperature readings.

MAX6675 IC has inbuilt cold-

junction compensation (that is, zero temperature reference) and it digitises the signal from a K-type thermocouple. It converts the signal to digital (binary) with an accuracy of 0.25°C. Temperature is displayed on four 7-segment displays (common-anode) without decimal point.

The basic principle of a thermocouple is that, if two junctions are exposed to two different temperatures, an emf is generated between the two junctions of two dissimilar metal

wires. K-type thermocouples can measure temperatures up to 1300°C, and are widely used in industries for measuring temperatures of furnaces, heaters and the like.

Circuit and working

The circuit diagram of the digital thermometer is shown in Fig. 1. It comprises ATmega16 MCU (IC2), MAX6675 converter (IC3), 5V voltage regulator 7805 (IC1), four 7-segment displays (DISP1-DISP4) and a few

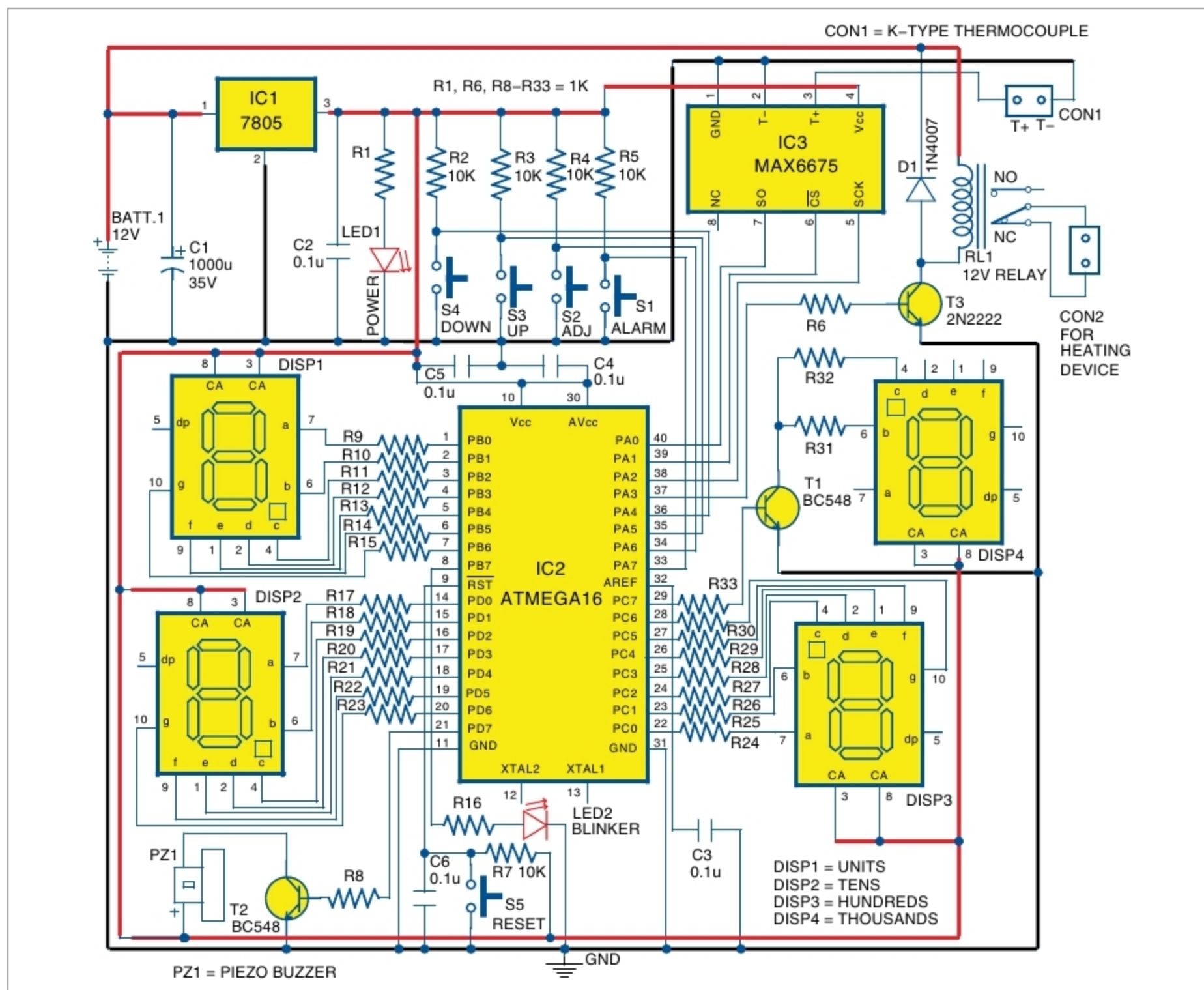


Fig. 1: Circuit diagram of digital thermometer